Assignment - 01

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Subject - operating system

* Types of OS

1] Batch OS

Definition - Executes jobs in batches without user interaction

Advantages:	Disadvantages
- High CPU utilization	- No direct use interac
- can process many jobs automatically	- long waiting time
- Suitable for repetitive	- Difficult to debug error.

2] Multiprogramming OS.

Defn - keeps multiples programs in memory and executes them one by one to keep the CPU busy.

Advantages	Disadvantages.
- Better CPU utilizatn	- Complex memory management
- Increases System efficiency	- Difficult Schedulı
- Reduces CPU idle time	- Requires more memory

Uses: - office computers, Software developr
Data processing System

3] Multitasking OS

Defn :- Allows a user to run multiple application at the same time.

Advantages	Disadvantages
- Improves productivity	- Requires more RAM
- Save time	- System may slow down many program
- Better user experienc	

4] Multiprocessing OS.

Defn: uses two or more processors to execute task simultaneously

Advantages:-	Disadvantages
- Fast processing	- Expensive hardware
- Higher reliability	- Complex system desig
- Better performance	- Higher power consumpt

5] Real time OS (RTOS)

Defn:- process task within a fixed time limit

advantages	Disadvantages
- very fast response	- Expensive develop
- Highly reliable	- limited flexibility
- Suitable for critical system.	- complex programmi

Uses:- Air traffic control system, medical equipm
- Industrial automation.

6] Distributed OS

Defn:- manages multiple connected computer as a single systems

advantages	Disadvantages
- Resource sharing	complex maintenance
- High performance	Network dependency
- Better reliability	Security challenges

uses:- cloud computing, Banking networks University and research network

7] Network OS (NOS)

defn:- manages computers connected over a network

advantages
centralized management
file, printer sharing.

Disadvantages:
Server dependency
maintenence cost.

Linux System cell.

Definition of System cell
A System call is a mechanism through which a user program request services from the operating system kernel. System calls provide an interface betn user application and the OS for performing operations.

Types of System calls.

1] Process control System calls

These System calls are used to create execute, terminate and manage process.

eg- "fork()", "execve()", "exit()", "wait()",

2] File management System cell.

These System calls are used to create open read, write, close and manipulate files.

eg- "open()", "read()", "write()", "close()", "lseek()", "Stat()".

3] Device management System calls.

These System calls allow communication with hardware devices such as disks printers, keyboards, network devices.

ex- "read" "write()" "fcntl()"

4] Memory management System calls.

These System calls manage memory allocation, dellocation, and mapping betn user process and memory.

ex- "mmap()", "brk()", "munmap()".

5] Information maintenance System call

These System calls obtain or modif System and process infon.

ex- "getPid()", "nice()", "getpriority()", "setpriority()",

6] communication System calls.

These System calls enable communica betn processes using pipes, Sockets, shared memory and message queues

ex- "pipe()", "Socket()", "Send()", "rec

Definitions of the Given System calls.

1] "Fork()"
→ Create a new child process by duplicating the calling (parent) process

2] "vFork()"
→ Create a child process that shares the parent's address space until it calls "execve()" or exits.

3] "clone()"
→ Creates a new process of thread with shared resources depending on specified flags.

4] "execve()"
→ Replace the current process image with a new program

5] "exit()"
→ Terminates a process after performing standard cleanup operations.

6] "_exite()"
→ Immediately terminates a process without performing standard library cleanup.

7] "wait()"
→ Suspends the parent process until one of its child processes terminates

8] "waitpid()"
→ wait for a specific child process to terminated

9] "waitid()"
→ wait for state change in child process and provides detailed status information.

10] "kill()"
→ send a singal to a process or process group

11] "getPid()"
→ Returns the process ID (PID) of the calling process

12] "getPPid()"
→ Returns the parents process ID (PPID) of the calling process

13] "nice()"
→ changes the sheduling priority of process

14] "setpriority()"
→ sets the scheduling priority of a process process group, or user.

15] "getpriority()"
→ Retrieves the scheduling priority of process

16] "open()"
→ opens an existing file or creates one if specified with appropriates flags

17] "create()"
→ creates a new file or truncates an existing file for writting

18] "close()"
→ close an open file descriptor

19] "read()"
→ Reads data from a file for device into a buffer

20] "write()"
→ writes data from a specified the file offset without changing the current file

21] "pread()"
→ writes data to a specified file offset without changing the current file pointer

22] "pwread()"
→ changes the current position of file pointer

23] "lseek()"
→ Retrieves information about file using its pathname

24] "stat()"
→ Retrieves information about a file using file descriptor.

25] "fstat()"
→ Retrieves information about a symbolic link without following it.

26] "lstat()"
→ checks whether a file exists and whether the process has permission to acess it

27] "acess()"
→ Deletes a files for the file system.

28] "unlink()"
→ changes the name or location of a file or directory

29] "rename()"
→ create a new directory

30] "mkdir()"
→ Removes an empty directory

31] "rmdir()"
→ duplicates an existing file descriptor

32] "dup()"
→ Duplicates a file descriptor to a specified descriptor number

33] "dup2()"
→ Permorms various operating of file descriptors such as duplication locking and changing flags.

35] "FSync()"
→ writes all modified file data and metadata to perment storage.

36] "fdatasync()"
→ writes modified file data to disk but may delay writing unecessary metadata

37] "truncate()"
→ changes the size of a file using the pathname

38] "Ftruncate()".
→ changes the size of a file using its file descriptor.